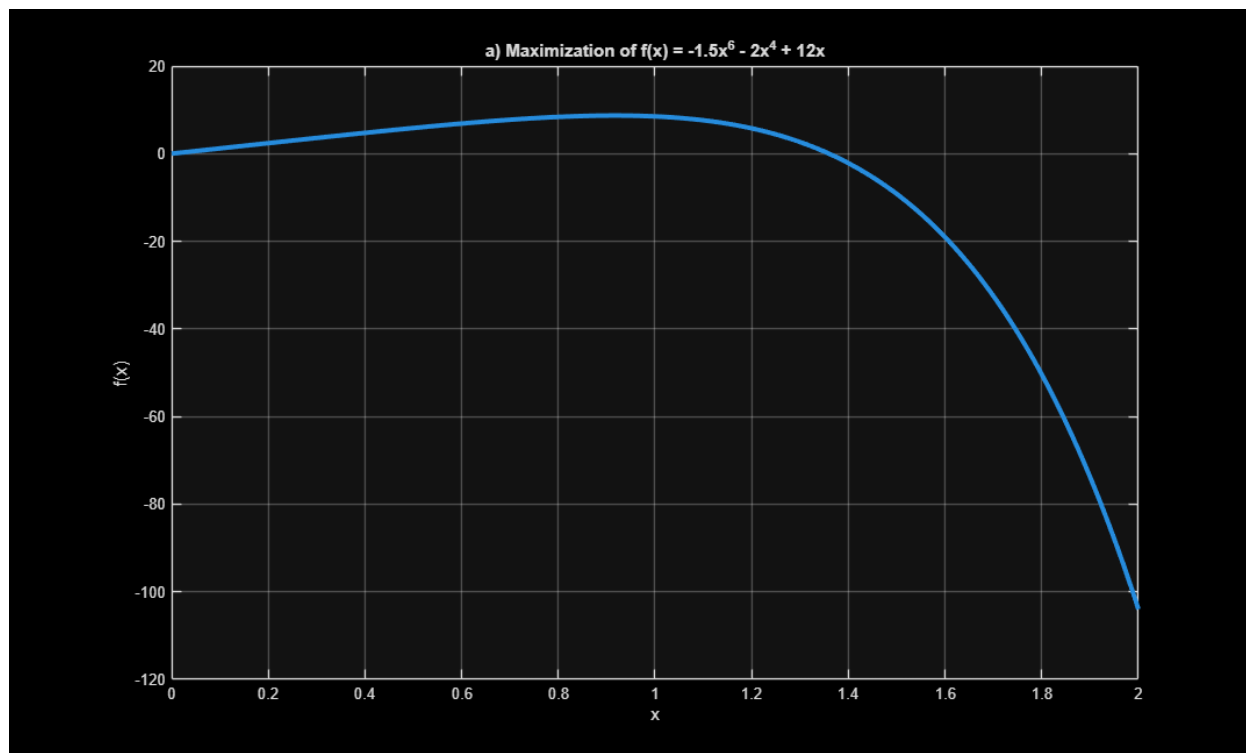

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```
function HW_2_5()  
  
    clear; clc; close all;  
  
    f = @(x) -1.5*x.^6 - 2*x.^4 + 12*x;  
    df = @(x) -9*x.^5 - 8*x.^3 + 12;
```

a)

```
figure('Name', 'Problem 5 Optimization', 'NumberTitle', 'off');  
x_plot = linspace(0, 2, 500);  
plot(x_plot, f(x_plot), 'LineWidth', 3);  
grid on; hold on;  
xlabel('x'); ylabel('f(x)');  
title('a) Maximization of f(x) = -1.5x^6 - 2x^4 + 12x');
```



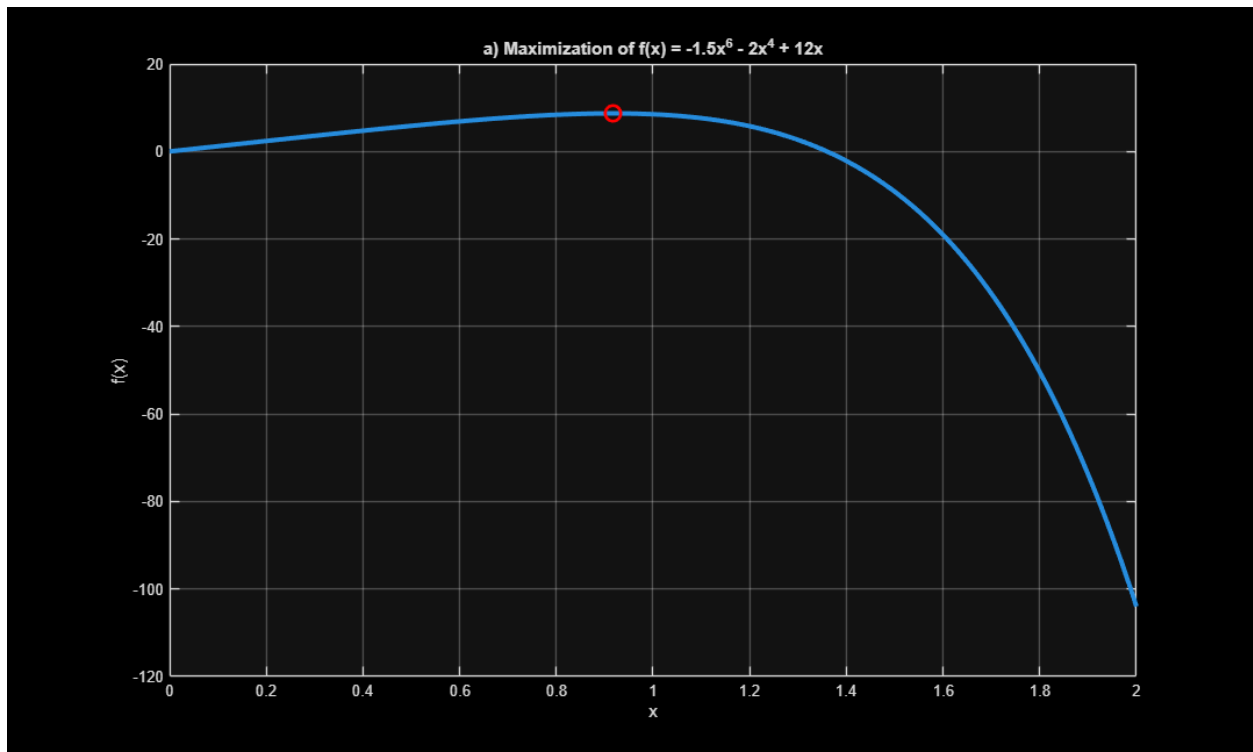
b)

```
x_fzero = fzero(df, 1);
max_fzero = f(x_fzero);

plot(x_fzero, max_fzero, 'ro', 'MarkerSize', 10, 'LineWidth', 2);

fprintf('=== (b) fzero Method ===\n');
fprintf('Optimal x: %.6f\n', x_fzero);
fprintf('Maximum f(x): %.6f\n\n', max_fzero);

=== (b) fzero Method ===
Optimal x: 0.916915
Maximum f(x): 8.697930
```



c)

```
x1 = 0; xu = 2;
tol = 1e-5;
[x_gold, max_gold] = golden_section_max(f, x1, xu, tol);

fprintf('=== (c) Golden-Section Search ===\n');
fprintf('Optimal x: %.6f\n', x_gold);
fprintf('Maximum f(x): %.6f\n\n', max_gold);
```

d)

```
x1 = 0; x2 = 1; x3 = 2;
[x_para, max_para] = parabolic_interpolation_max(f, x1, x2, x3, tol);

fprintf('=== (d) Parabolic Interpolation ===\n');
fprintf('Optimal x: %.6f\n', x_para);
fprintf('Maximum f(x): %.6f\n\n', max_para);

end
```

Golden section search for c)

```
function [x_opt, f_opt] = golden_section_max(f, x1, xu, tol)
    R = (sqrt(5) - 1) / 2; % Golden ratio (~0.61803)
    d = R * (xu - x1);
    x1 = x1 + d;
    x2 = xu - d;
    f1 = f(x1);
    f2 = f(x2);

    while (xu - x1) > tol
        if f1 > f2
            x1 = x2;
            x2 = x1;
            f2 = f1;
            x1 = x1 + R * (xu - x1);
            f1 = f(x1);
        else
            xu = x1;
            x1 = x2;
            f1 = f2;
            x2 = xu - R * (xu - x1);
            f2 = f(x2);
        end
    end
    x_opt = (x1 + xu) / 2;
    f_opt = f(x_opt);
end
```

```
=== (c) Golden-Section Search ===
Optimal x: 0.916914
Maximum f(x): 8.697930
```

Parabolic interpolation optimization algorithm for d)

```
function [x4, f4] = parabolic_interpolation_max(f, x1, x2, x3, tol)
    max_iter = 100;
    x4 = x2;
```

```

for iter = 1:max_iter
    f1 = f(x1); f2 = f(x2); f3 = f(x3);

    num = f1*(x2^2 - x3^2) + f2*(x3^2 - x1^2) + f3*(x1^2 - x2^2);
    den = 2*f1*(x2 - x3) + 2*f2*(x3 - x1) + 2*f3*(x1 - x2);

    if abs(den) < 1e-12, break; end
    x4_new = num / den;

    if abs(x4_new - x4) < tol
        x4 = x4_new;
        break;
    end
    x4 = x4_new;

    if x4 > x2
        if f(x4) > f(x2)
            x1 = x2; x2 = x4;
        else
            x3 = x4;
        end
    else
        if f(x4) > f(x2)
            x3 = x2; x2 = x4;
        else
            x1 = x4;
        end
    end
end
f4 = f(x4);
end

=== (d) Parabolic Interpolation ===
Optimal x: 0.916914
Maximum f(x): 8.697930

```

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